

RitualResonate: Embodied Haptic Interaction with Chinese Chime Bells Bianzhong

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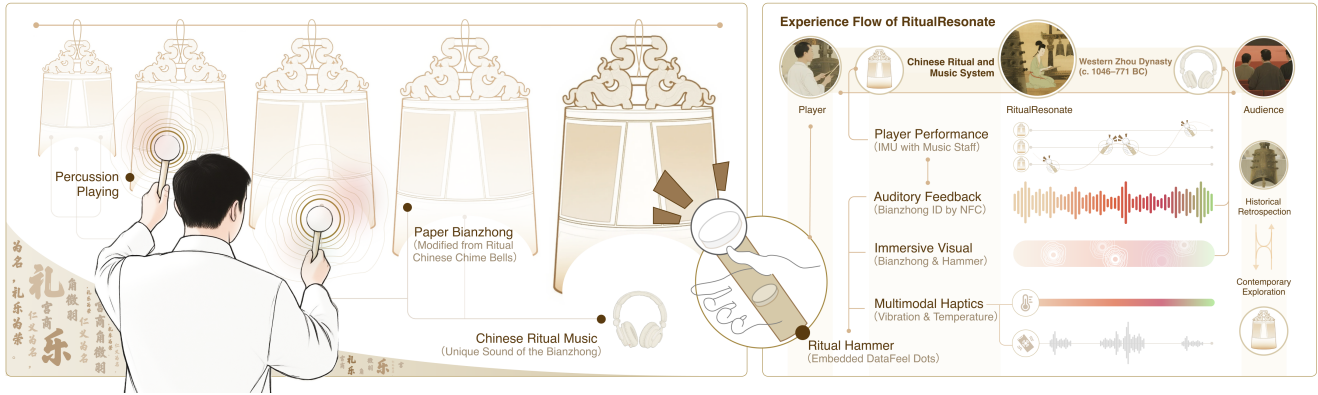


Figure 1: Project Overview and Operation Flow Diagram

Abstract

This study presents RitualResonate, an interactive device centered around a "hammer" as the core element, integrating the cultural imagery of chime bells and multi-modal tactile feedback. The system collects real-time data on striking actions and generates multi-channel feedback experiences, exploring the possibility of enhancing cultural rituals through tactile sensations, and providing a new path for the digital translation of traditional rituals and music.

CCS Concepts

• Human-centered computing → Interaction design; • Applied computing → Arts and humanities.

Keywords

Multi-modal haptics, cultural heritage, embodied interaction, Chinese traditional culture, ritual music

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1 Introduction

Since the mid-nineteenth century, museum exhibitions have largely followed a "do not touch" policy, underscoring the priority given to artifact preservation and the authority of scholarly interpretation. The digitization of cultural heritage has prompted a rethinking of multi-sensory exhibition formats. An increasing number of cultural heritage artifacts are now presented in public spaces and interactive installations through images, audio, and three-dimensional models. This transformation turns fragile and valuable artifacts into data-driven elements, and the potential of haptic interaction in cultural experiences is increasingly recognized [4].

Originating during the Zhou dynasty and flourishing through the Spring and Autumn, Warring States, and into the Qin and Han periods, the bianzhong (chime bells) represent one of the most iconic ritual instruments in ancient Chinese music. They are widely recognized for their solemn appearance and lingering resonance. Cast in bronze, each bianzhong set consists of multiple bells of varying sizes, carefully arranged on wooden frames according to their pitch. Each bell produces a distinct tone when struck [6, 9]. Compared to other percussive instruments, bianzhong produces an unusually rich and prolonged vibratory response. Its low-frequency resonance and nuanced overtone structures allow for a highly differentiated tactile experience, where pitch variations and timbral subtleties can be physically perceived through vibration. The complex decay profiles of different bells create a multi-sensory texture, bridging sound, material, and embodied perception. However, as highly ceremonial and historically significant instruments, bianzhong often remain distant from everyday experience. Public engagement is largely restricted to passive forms such as watching or listening to staged performances, with few opportunities for direct physical engagement or participation. As a result, audiences are rarely able to develop embodied understanding or meaningful emotional resonance with the instrument.

Existing research has identified both music and touch as intrinsic modalities of emotional expression in human culture. Their integration has been shown to elicit notable physiological and psychological responses, enhancing user experience and potentially enabling novel forms of communicative mapping. The perceptual qualities of haptic feedback often depend on variables such as frequency, amplitude, duration, material, and the location on the body. These parameters exhibit cross-modal correspondences with musical features including pitch and tempo [1]. Current music interaction systems are largely utilitarian in orientation, focusing on rhythm training and signal-based prompting, particularly for modern percussion instruments [3, 5, 8]. In virtual music systems, traditional instruments are often digitally reconstructed—such as virtual chime bells or digital guqin—emphasizing visual realism and high-fidelity sound reproduction. However, the potential of haptic interaction to enrich cultural experience remains underexplored, limiting the system’s capacity to convey nuanced expressions of bodily rhythm and cultural meaning [2, 7, 10].

Building on this premise, we introduce RitualResonate—an interactive installation inspired by the cultural symbolism of ancient Chinese chime bells, centered on a handheld “hammer” that integrates lightweight striking gestures with multimodal haptic feedback. Rather than replicating traditional instruments, our focus lies in reimagining the ancient bianzhong through contemporary design interpretations.

2 System Design

RitualResonate consists of a set of lightweight, foldable “paper-chime” components and two handheld “hammers” embedded with sensing and feedback modules. By striking the paper chimes with both hammers, the system captures real-time motion parameters (including speed, frequency, and contact duration) and translates them into synchronized visual, auditory, and haptic responses, creating an embodied and perceivable cultural interaction. Unlike traditional

musical interfaces that focus solely on auditory reproduction, RitualResonate places greater emphasis on the hammer’s complex, multimodal tactile feedback system—namely, a layered mapping of bell sound, visual cues, vibration, and temperature feedback, which together form an integrated sensory response.

- **Vibration feedback** simulates resonance through adjustable amplitude, frequency, and duration, such as low-frequency strikes yield deep and sustained tremors, while high-frequency taps produce short, rapid pulses;
- **Thermal feedback** reflects the emotional tone or ritual setting of the performance, such as the solemnity of ancient court ceremonies or the warmth of festive scenes—with temperature level and transition speed modulated by striking frequency or musical character;
- **Visual and auditory feedback** employs LED color and rhythmic sound effects to visualize tempo and spatial variation.

To heighten the perceptual contrast between material and meaning, we adopt paper as the structural medium for chimes. This choice embodies not only physical qualities—lightweight, foldable, and easily assembled, but also symbolizes the accessibility and reconfigurability of cultural heritage. By reinterpreting the traditionally solemn and weighty bianzhong through ephemeral materials, PastPulse creates a compelling dialogic tension: preserving ritualistic visual cues while inviting users to engage with tradition through a contemporary, embodied lens.

3 Implementation

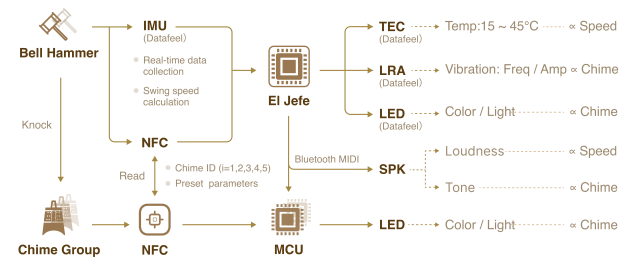


Figure 2: Hardware System Architecture Overview

RitualResonate makes full use of the DataFeel DevKit provided by the UIST Student Innovation Contest. Building upon this foundation, as shown in Figure 2, we integrated a range of additional modules (including a speaker unit, LED breathing lights, micro heating elements, multiple haptic motors, and accelerometers) and employed fabrication techniques such as 3D printing. To construct a minimal yet effective cultural interaction device, all modules were embedded within the dual-hammer structure, positioning the hammers as the central nodes of the user experience. Each sensor and actuator was strategically placed within the handle and head of the hammers to optimize responsiveness and tactile clarity.

To progressively guide users in understanding the connection between chime-bell culture and haptic feedback, we developed a three-step interaction sequence. First, users are introduced to the

cultural imagery of bianzhong through audiovisual cues and invited to perform initial tapping actions on bell surfaces made of different materials. Second, they engage in guided rhythmic playing based on simple notation prompts, experiencing the dynamic mapping of motion with vibration, sound, and temperature feedback. Finally, users may either perform a short 20-second traditional rhythmic segment or create their own sequences, using haptic responses to generate personalized expressions of cultural resonance. Participants may engage as performers in immersive interaction or as observers enjoying a multisensory presentation. The system is also well-suited for public exhibition and cultural outreach.

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